Deep Ensembles — what applies to our reconstruction

Lakshminarayanan, Pritzel & Blundell, NeurIPS 2017 (arXiv:1612.01474)

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Where we start: the model has no uncertainty to report

- Our 4DVarNet is trained on 4DVarNet's Eq. 14 — the plain squared error between the reconstruction and the truth.
- Squared error constrains the MEAN and nothing else. There is no second output, and no term in the objective that any second output could have been fitted against.
- So the gap is not that we have not computed the uncertainty yet. It is that a model trained this way has no quantity that represents it — which is why it cannot be added as a post-processing step, and why the change has to reach the training objective.
- What follows is what the paper offers for exactly this situation, and how much of our existing model it would disturb.

What the paper proposes instead (their Eq. 1)

- The network emits two values per output — a mean and a variance — and the target is treated as a draw from N(μ, σ²).

$$-\log p_\theta(y_n \,|\, x_n) \;=\; \frac{\log \sigma_\theta^2(x)}{2} \;+\; \frac{(y-\mu_\theta(x))^2}{2\,\sigma_\theta^2(x)} \;+\; \text{constant}$$

grows with σ:
punishes claiming ignorance

shrinks with σ:
punishes confident errors

minimised at $\sigma_\theta^2(x) = (y-\mu_\theta(x))^2$ — the predicted variance is fitted to the actual squared error

- σ² is kept positive with a softplus and floored at a small minimum variance for numerical stability.
- It is heteroscedastic: σ² is a function of the input, so the model can be confident in some cells and not in others, instead of quoting one error bar for the whole field.

The second thing worth taking: what the uncertainty is made of

At test time the ensemble is a uniformly-weighted mixture:

$$p(y|x) \; = \; M^{-1} \sum_{m=1}^{M} p_{\theta_m}(y \,|\, x, \theta_m)$$

approximated by a single Gaussian with the mixture's first two moments:

$$\mu_*(x) = M^{-1} \sum_m \mu_{\theta_m}(x) \qquad \sigma_*^2(x) = M^{-1} \sum_m \left(\sigma_{\theta_m}^2(x) + \mu_{\theta_m}^2(x) \right) - \mu_*^2(x)$$

$$\text{equivalently} \quad \sigma_*^2 \; = \; \underbrace{M^{-1} \sum_m \sigma_{\theta_m}^2}_{\text{aleatoric}} \; + \; \underbrace{\mathrm{var}_m(\mu_{\theta_m})}_{\text{epistemic}}$$

- First term: the average of what each network itself reports it cannot account for. Adding members does not reduce it. Second term: how far the networks' means are from each other on the same input — it shrinks as they come to agree.

What we would not take, and what it costs

- Not taken — adversarial training (their Sec. 2.3)
  - it is step 2 of their recipe but optional, and their Table 2 states it does not significantly help on the regression benchmarks
  - its input perturbation, a step along the sign of the loss gradient, also has no clear reading on an observation field: our inputs are a measured state and a mask, not free variables
- What it costs
  - M independent trainings and M forward passes at inference; the paper recommends M = 5 and reports diminishing returns beyond it
  - accuracy: the objective optimises NLL rather than squared error